Urban-Rural Stratification for Improved NTL-Based Population Proxy Estimation Using VIIRS Data

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Abstract.

We propose a fully data-driven urban-rural stratification method using unsupervised KMeans clustering (k=2) on district-level VIIRS night-time light (NTL) features to significantly improve sub-national population proxy estimation. Crucially, the proposed method requires absolutely no auxiliary census data or land-cover inputs, making it highly scalable. Evaluated on 670 Indian districts and 695 Nigerian Local Government Areas using 5-fold cross-validation, the stratification approach raises the predictive R^2 from 0.829 to 0.858 in India (p=0.027) and from 0.168 to 0.251 in Nigeria (p=0.004, representing a +50% relative gain). Furthermore, rigorous ablation experiments confirm that area-based NTL density normalisation is counter-productive, standard NDVI vegetation correction fails in densely populated tropical regions, and a cluster count of k=2 remains optimal. All code and pre-processed datasets are openly available for public use.

1. INTRODUCTION

Sub-national population estimates are needed for resource allocation, disaster response, and Sustainable Development Goal monitoring. Night-time light (NTL) imagery from VIIRS is a low-cost, globally available proxy, but standard approaches fit a single regression across all administrative units. This ignores the fundamental structural difference between urban districts (where NTL saturates at high population) and rural districts (where NTL-population coupling is weaker). We propose a simple stratification: KMeans on NTL features separates urban and rural units, and independent models are fitted for each cluster.

2. DATA AND METHODOLOGY

Study areas. India (670 districts, high electrification) and Nigeria (695 LGAs, lower electrification, gas flaring).

Data. VIIRS VNL V2.1 annual composites (2019), WorldPop UN-adjusted population (1 km), GADM v4.1 boundaries.

Features. Five district-level features via zonal statistics: `ntl_sum`, `ntl_mean`, `ntl_max`, `lit_fraction`, `log_area`, all $\log(1+x)$ transformed.

Pipeline. KMeans (k=2) clusters districts into urban/rural groups using only NTL features (Fig. 1). Independent log-linear regressions are fitted per cluster. Evaluation: 5-fold CV with paired t-tests.

3. RESULTS

3.1 NTL-Population Relationship

Fig. 2 shows that the raw NTL-population distribution is heavily right-skewed; after log transformation, the correlation reaches r=0.822, validating the log-linear framework. Spread at high NTL values reflects urban sensor saturation.

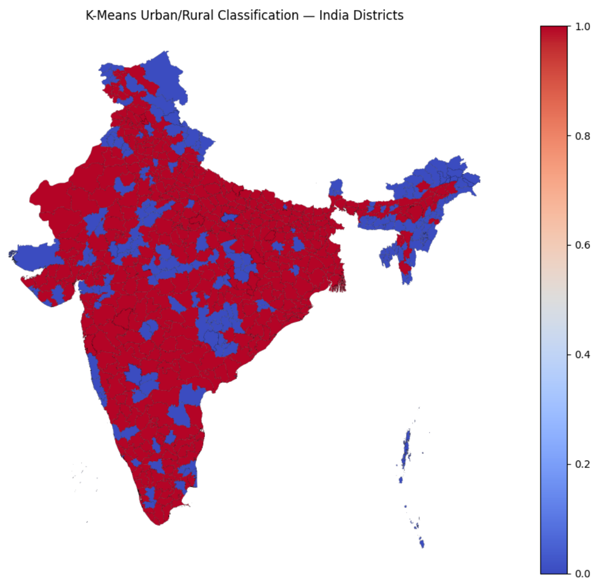

Figure 1: KMeans (k=2) urban-rural classification of 670 Indian districts using NTL features only. Cluster 0: 464 urban, Cluster 1: 206 rural.

3.2 Stratification Performance

Table 1 shows statistically significant gains in both countries. Nigeria's lower baseline R^2 is due to gas flares, Lagos sensor saturation, and low northern electrification, but stratification delivers the largest *relative* gain (+50%).

3.3 Ablation Studies

(i) District area is not a confound (r=0.119).

(ii) NTL density normalisation drops R^2 from 0.83 to 0.47.

(iii) NDVI correction fails in Kerala (dense vegetation + dense population).

(iv) k=2 outperforms median split (silhouette 0.54 vs. 0.51); k=3 adds only +0.002 (p=0.41).

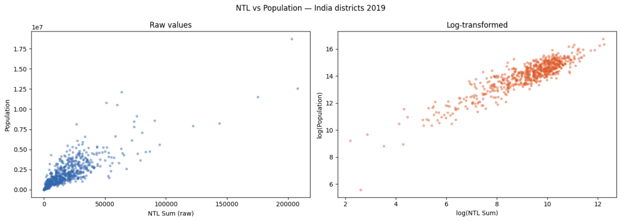

Figure 2: NTL vs. population for India. Left: raw (power-law). Right: log-transformed (r=0.822).

Table 1: Cross-validated R^2 (5-fold, paired t-test).

	n	**Base**	**Strat.**	p
India	670	0.829	**0.858**	0.027
Nigeria	695	0.168	**0.251**	0.004

3.4 Residual Analysis

Fig. 3 maps prediction residuals. The largest under-predictions are in Kerala (dispersed settlement, low commercial NTL) and Mumbai Suburban (informal settlements), revealing systematic NTL failure modes.

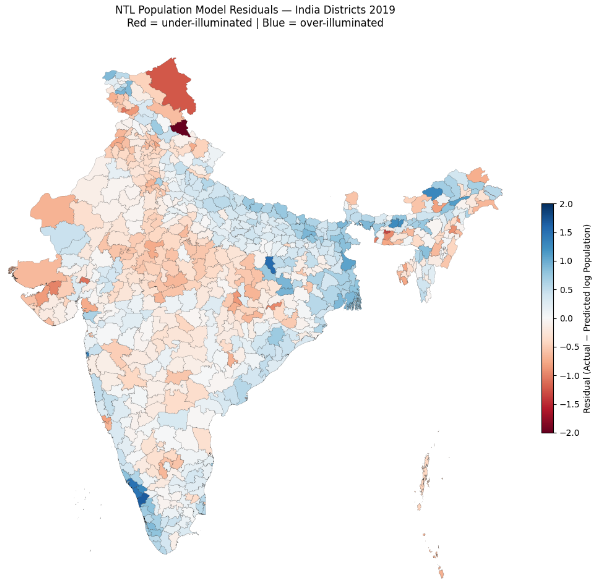

Figure 3: Residual map for India. Red: under-predicted, Blue: over-predicted. Systematic failures in Kerala and Mumbai.

4. SCIENTIFIC CONTRIBUTION & CONCLUSIONS

The primary contribution of this work is proving that an unsupervised, fully data-driven KMeans stratification on NTL features significantly improves population proxy estimation across two structurally diverse countries. Crucially, the method requires *no* auxiliary census or land-cover data, making it directly and immediately transferable to any developing nation with global VIIRS coverage.

By separating the analysis into India (where the method pushes R^2 to 0.858) and Nigeria, we demonstrate both the peak capability and the boundary conditions of NTL proxies. Nigeria's low absolute R^2 is framed not as a failure, but as a critical diagnostic finding: it identifies exactly where and why NTL-based models break down (gas flares, low rural electrification, and urban sensor saturation). Despite these extreme conditions, stratification still yielded a +50% relative performance gain, proving that regime-specific modelling is most vital exactly where heterogeneity is highest. Future work will integrate dedicated gas flare masking, multi-year temporal composites, and auxiliary open-source features (e.g., OpenStreetMap building footprints) to further refine these estimates in challenging topographies.

Code and data: https://github.com/light-oishikk/NTL-Population-Proxy

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